ShadowIntern: A Multi-Agent Framework for Company-Specific Engineering Culture Extraction and Technical Interview Preparation

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Abstract

Technical interview preparation platforms today offer generic problem sets and scripted mock interviews that lack company-specific context. We propose ***ShadowIntern****, a multi-agent LLM framework that extracts a target company's engineering culture from publicly available data sources — including GitHub repositories, technical blog posts, job descriptions, and interview review platforms — and uses this extracted "Company DNA" to generate contextually grounded intern-level tasks, simulate company-calibrated code review, and conduct mock interviews using company-specific question patterns. The system employs five specialised agents operating in a sequential pipeline. Over a four-week shadow period, candidates are progressively trained on problems that mirror the scale, stack, and engineering values of a specific target company. We describe the system architecture, the Company DNA extraction methodology, and a proposed evaluation framework for measuring preparation effectiveness.*

Keywords: multi-agent systems, LLM, interview preparation, engineering culture modelling, company-specific task generation, AI-powered mentorship

1. Introduction

The technical hiring process at leading technology companies rewards candidates who not only possess algorithmic knowledge but also demonstrate familiarity with company-specific engineering culture — the stack, the scale, the values, and the problem domains that define a company's day-to-day work. Existing preparation platforms such as LeetCode, HackerRank, and Pramp address generic algorithmic skill development but offer no mechanism for developing company-specific cultural fluency.

We identify three key gaps in current approaches: (1) *context blindness* — preparation is decoupled from a specific company's actual engineering problems; (2) *culture opacity* — candidates have no structured way to internalise how a particular company thinks about software; and (3) *feedback miscalibration* — code review and mock interviews are not grounded in the standards of a specific target organisation.

We propose ShadowIntern, a five-agent LLM pipeline that addresses all three gaps by constructing a "Company DNA" profile from public data and using it to simulate a four-week intern-level work experience at the target company, without requiring any internal access or industry connections.

2. Related Work

Mock Interview Systems. MockLLM [1] proposes a multi-agent framework for person-job matching via simulated interview conversations between LLM-played interviewers and candidates. While MockLLM focuses on resume-to-job fit evaluation, ShadowIntern focuses on a longitudinal, culture-immersive preparation process rather than a single-shot interview simulation.

Agent-Based Workplace Benchmarks. TheAgentCompany [2] benchmarks LLM agents on real workplace tasks, demonstrating that frontier models can autonomously complete approximately 30% of realistic work tasks. ShadowIntern inverts this paradigm: rather than evaluating agents on work tasks, it uses agents to *train humans* on company-specific work tasks.

Multi-Agent Software Simulation. MetaGPT [3] and ChatDev simulate software development organisations using specialised LLM agents. ShadowIntern draws on this principle of role-specific agents but applies it to the educational domain of candidate preparation.

3. System Architecture

ShadowIntern comprises five specialised agents operating in a sequential pipeline, illustrated below. Each agent's output directly conditions the behaviour of the next.

Agent	Input	Output
1. Company Intelligence	Company name	CompanyDNA object
2. Culture Modeller	CompanyDNA	EngineerProfile persona
3. Task Generator	EngineerProfile + week no.	Weekly task assignments
4. Shadow Mentor	Student submission + profile	Severity-tagged review
5. Interview Simulator	Profile + 4-week history	Hire verdict + gap report

Table 1: ShadowIntern five-agent pipeline overview.

Company Intelligence Agent. Given a company name, this agent aggregates data from multiple public sources: GitHub organisation repositories (coding style, dominant stack, PR patterns), official engineering blogs (architectural decisions, scale narratives), job descriptions from the past three years (required skills, values language), and interview review platforms (actual questions asked, process structure). The output is a structured CompanyDNA object encoding stack, scale context, known problem domains, and culture signals.

Culture Modeller. This agent transforms the CompanyDNA into an EngineerProfile: a prompt-ready senior engineer persona calibrated to the specific company. The profile encodes what the company values (e.g. idempotency-first design at Razorpay), what it rejects (e.g. insufficient observability), and at what level it operates (e.g. 5M webhooks/day scale context).

Task Generator. Weekly tasks are generated in a progressive difficulty curve: component-level (Week 1) through full system design with failure modes (Week 4). Tasks are explicitly grounded in the company's known problem domains and constrained by its actual scale and SLA requirements.

Shadow Mentor. Student submissions are reviewed by an LLM agent conditioned on the EngineerProfile. Feedback is severity-tagged (P0/P1/P2 in payments terminology for fintech companies) and framed as "a [Company] engineer would reject this because...", making the cultural grounding explicit. Tasks with P0 issues are returned for revision before the student advances.

Interview Simulator. After four weeks, a three-round mock interview is conducted: algorithmic (company-calibrated difficulty), system design (drawn from the company's known interview question bank), and behavioural (company values signals). A hire/no-hire verdict and gap analysis are returned.

4. Discussion and Future Work

ShadowIntern introduces the notion of *company-specific culture immersion* as a preparation modality distinct from algorithmic practice or generic mock interviews. The key hypothesis is that candidates who have internalised a company's engineering values and problem patterns will demonstrate measurably stronger contextual alignment in actual interviews.

Several limitations warrant acknowledgement. First, the quality of the Company DNA is bounded by the richness and accuracy of publicly available data; companies with sparse public engineering footprints will yield less precise profiles. Second, the system cannot replicate internal context (undocumented conventions, team-specific practices). Third, the scraping layer involves data sources with varying terms of service, which must be addressed before any production deployment.

Future work will focus on: (1) a formal user study comparing ShadowIntern-trained candidates against a control group using existing platforms, with interview performance as the primary outcome metric; (2) automated evaluation of Company DNA quality against ground-truth profiles provided by verified employees; and (3) exploration of a B2B model where companies co-author the task bank, improving fidelity while creating a commercial incentive for adoption.

References

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